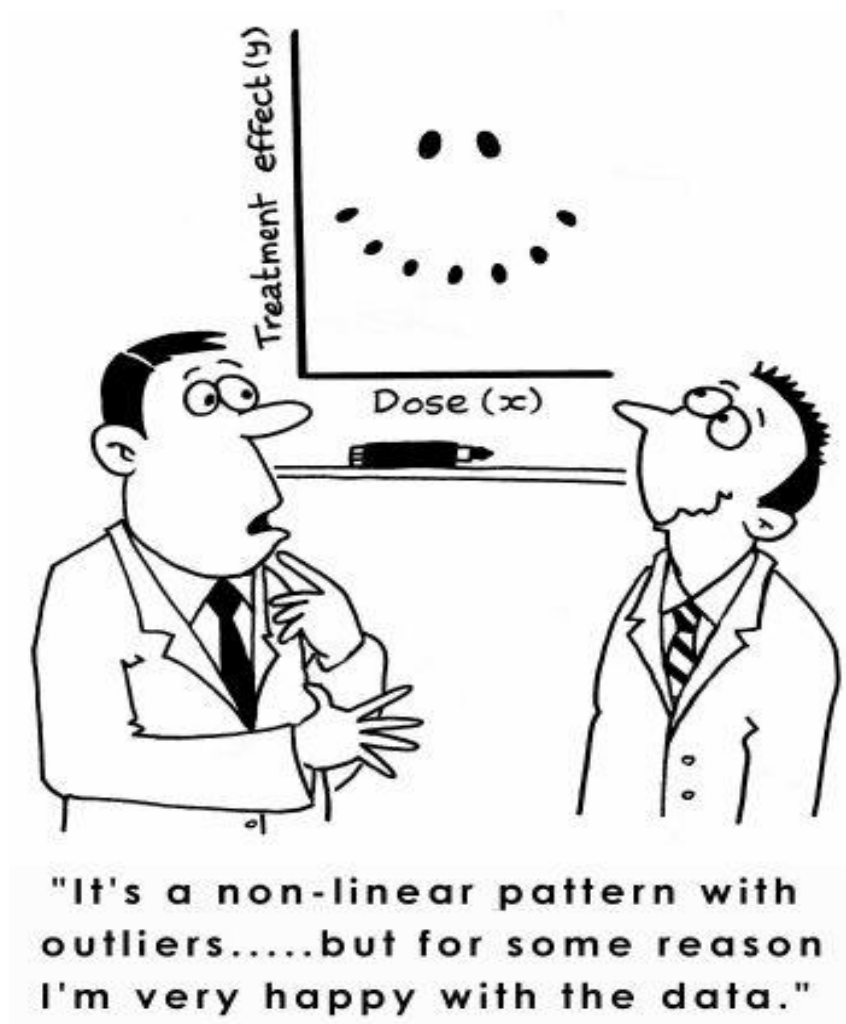


Data Preparation Session 6 Handout

Outliers



In our pipeline, we arrived to import the data, check and fix its general structure, and arrange columns in the right way. For a while, we focus on each column one by one. At first, we looked for missing values, followed by duplicates, and now we work with outliers.

Outliers are data points distant from the norm, or in other terms, unusual observations. Such a value can be an error in the database but can be correct and be true outlier. The population of Paris in France, or China in the world are far from other observations, but they are part of the dataset.

What to do with them?

1. **Detect** them using, for example uni- and bivariate descriptive statistics, clustering.
2. If possible, **figure out** (use other sources, simple logic) to reconstruct the correct value.
3. If possible, estimate it: impute a possible and likely value (the mean, the median, the mean of a group, etc.), cap it (use a reasonable maximum value for very large or minimum for very small values), predict with models (e.g. regression models can be used).
3. **Remove the value**, and replace it by NA.
4. **Remove the observation**: the complete row will be eliminated. Use it, if with an NA, the observation is useless or can distort/bias the whole analysis.

How to detect them?

After loading data, we have to check it to understand it and to eliminate problems.

```
dataHR <- read_excel("dataHR_labeled.xlsx")
```

The first 6 rows are given by head():

```
head(dataHR)
```

The last 6 rows are given by tail():

```
tail(dataHR)
```

A summary of descriptive statistics that works fine for numeric variables and factors.

```
summary(dataHR)
```

Filtering and subsetting is possible:

```
summary(dataHR[dataHR$wage>5000,])
```

Frequency distribution for one variable that works really fine for character variables:

```
table(dataHR$sex)
```

A boxplot (box-and-whiskers plot) can be useful.

```
boxplot(dataHR$sage)
```

Frequency distribution (cross-tabulation) for two variables:

```
table(dataHR$sex, dataHR$edu)
```

Proportion (percentages) for a table (% of grand total, % of row total, % of column total):

```
prop.table(table(dataHR$sex, dataHR$edu))
```

```
prop.table(table(dataHR$sex, dataHR$edu), 1)
```

```
prop.table(table(dataHR$sex, dataHR$edu), 2)
```

Finally, some fun 😊

```
tblFun <- function(x){
```

```
  tbl <- table(x)
```

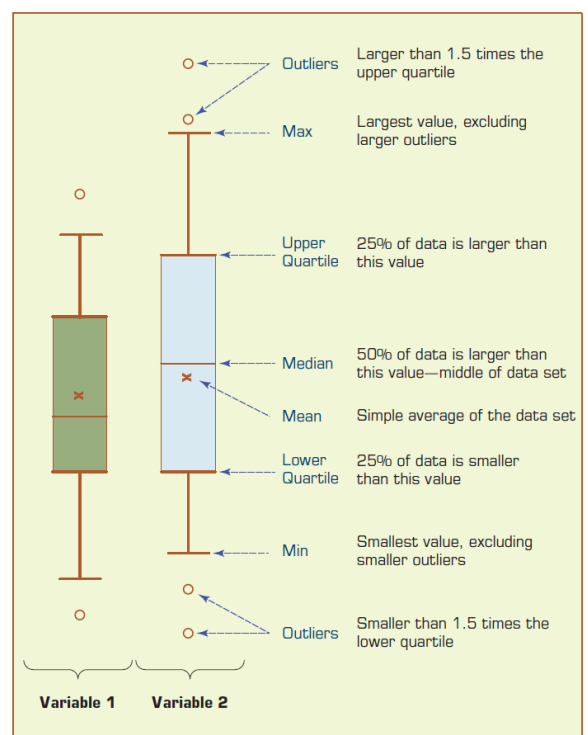
```
  res <- cbind(tbl, round(prop.table(tbl)*100,2))
```

```
  colnames(res) <- c('Count', 'Percentage')
```

```
  res
```

```
}
```

```
tblFun(dataHR$sex)
```



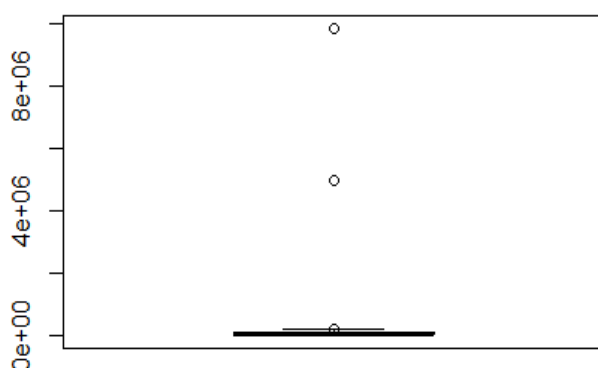
In the first exercise, we look for evidently extreme values. The data is about salaries of USA White House employees.

```
library(tidyverse)
```

```
whitehouse <- read_csv("whitehouse.csv", col_types="cnccci")
```

Let's check the salary variable (pay attention to spelling, the variable name is Salary and not salary).

```
boxplot(whitehouse$Salary)
```



We have two huge values around 5 and 9 million dollars, everything else seems to be below 1 million dollars. Let's find them. We filter for them:

```
whitehouse %>% filter(Salary>1000000)
```

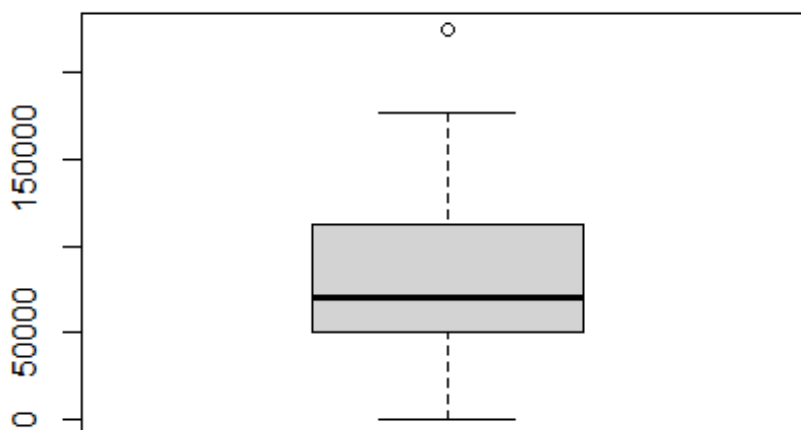
```
## # A tibble: 2 × 6
##   Name                Status  Salary `Pay Basis` `Position Title`  Year
##   <chr>              <chr>   <dbl> <chr>      <chr>          <int>
## 1 Case, Michael A.   Employee 5000000 Per Annum    SENIOR WRITER     2011
## 2 Blair, Patricia A. Employee 9866900 Per Annum    CHIEF CALLIGRAPHER 2015
```

None of them seem to be correct values. If we cannot check the real values, we set them missing (NA).

```
whitehouse <- whitehouse %>%
  mutate(Salary=ifelse(Salary>1000000, NA, Salary))
```

Let's check again the salary variable.

```
boxplot(whitehouse$Salary)
```



We have one outlier, a bit over 200,000 dollars. Let's find this person.

```
whitehouse %>% filter(Salary>200000)
```

```
## # A tibble: 1 × 6
##   Name                Status   Salary `Pay Basis` `Position Title`   Year
##   <chr>              <chr>   <dbl> <chr>      <chr>              <int>
## 1 Wheeler, Seth F. Detailee 225000 Per Annum  SENIOR POLICY ADVISOR 2013
```

A Senior Policy Advisor is an important position, we may believe that it's a correct value. You can check it here: <https://www.reuters.com/article/usa-obama-salaries/who-makes-the-most-in-obamas-white-house-idUSL2N0F41WU20130628/>

Back to school. We will check students' test scores in a given primary school.

The database has information about test scores of 217 students. The variables in the database:

Variable	Content	Possible values
studentID	The ID number of the student	10001, 10002, 10003, ... 10217
age	The age of the student in years	The school starting age can be 5 or 6 years (typically 6, but 5 is possible)
grade	School grade, students start at grade 1, when finishing, they move to grade 2, and so on, normally every year they move one grade up.	1, 2, ..., 8 If a student started the school at the normal age 6 in grade 1, they will be at age of 10 in grade 5 and at age of 13 in grade 8.
testScore	Score on a mathematics test of 120 points	0-120

```
tests <- read_csv("testscores.csv")
```

```
## Rows: 217 Columns: 4
## — Column specification —————
## Delimiter: ","
## dbl (4): studentID, age, grade, testScore
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

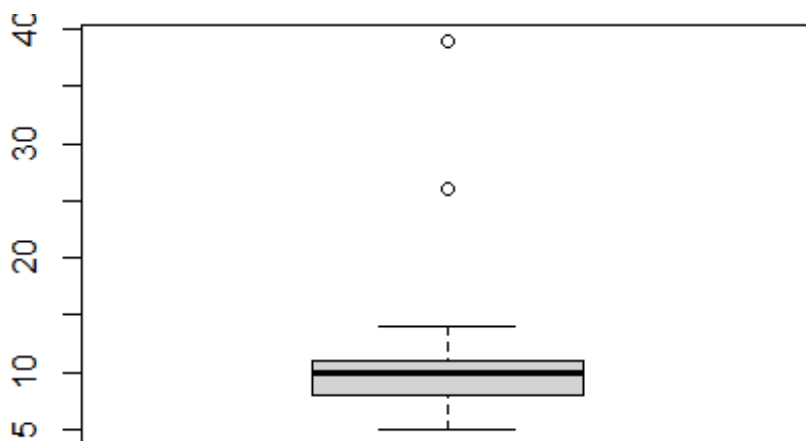
Let's check the summary statistics. It can help detecting very low and very large values.

```
summary(tests)
```

```
##   studentID      age      grade      testScore
##   Min.   :10001   Min.    : 5.000   Min.    :1.0   Min.    : 59.00
##   1st Qu.:10055   1st Qu.: 8.000   1st Qu.:3.0   1st Qu.: 75.00
##   Median :10109   Median :10.000   Median :4.0   Median : 83.00
##   Mean   :10109   Mean    : 9.871   Mean    :4.3   Mean    : 82.43
##   3rd Qu.:10163   3rd Qu.:11.000   3rd Qu.:6.0   3rd Qu.: 89.00
##   Max.    :10217   Max.    :39.000   Max.    :8.0   Max.    :114.00
```

The student ID is just an ID, nothing to see. The age is surely not OK, a 39 old pupil in the primary school? The grade is as in the description, the test score is also acceptable.

```
boxplot(tests$age)
```



We have not only one, but two outliers. Look at them.

```
tests %>% filter(age>15)
```

```
## # A tibble: 2 × 4
##   studentID   age grade testScore
##   <dbl> <dbl> <dbl>     <dbl>
## 1    10115    39     2         91
## 2    10116    26     7         70
```

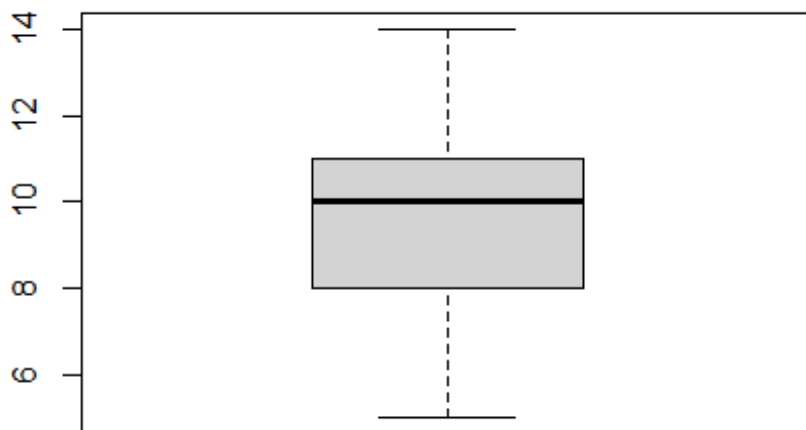
We have a few possibilities. We can set their age to NA, as the given one is almost surely erroneous. We can call the school to ask for additional information; Or we can use a simple model. We suppose that their grade is correct, and we know that a typical pupil starts the school at the age of 6, so grade 1 is at 6, grade 2 at 7, ... grade 7 at 12.

Do mutate the dataset.

```
tests <- tests %>%
  mutate(age=ifelse(studentID==10115, 7, age)) %>%
  mutate(age=ifelse(studentID==10116, 12, age))
```

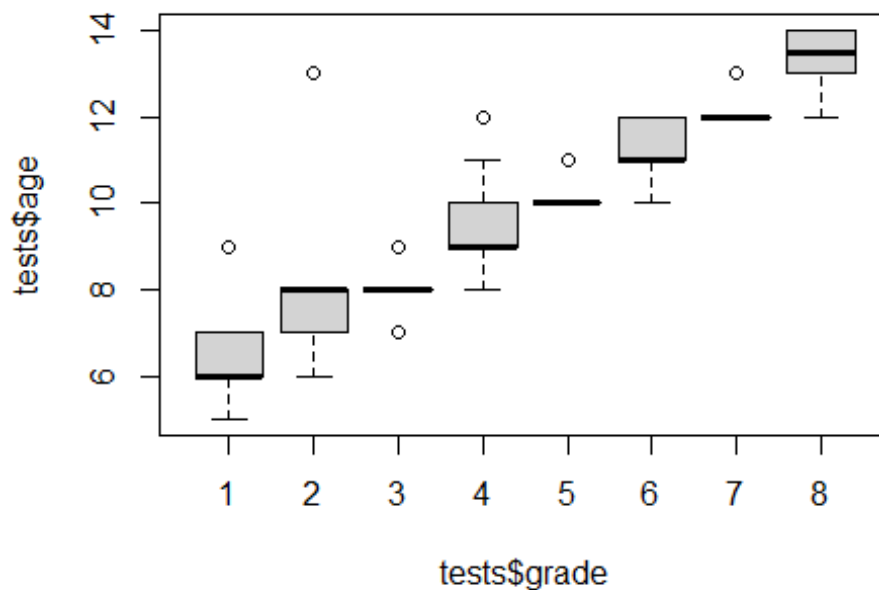
The boxplot with the clean data:

```
boxplot(tests$age)
```



Before we think that everything is solved; we can check a bivariate chart.

```
# But Look at it by grade level  
boxplot(tests$age~tests$grade)
```



A pupil in first grade at the age of 9, a pupil in grade 2 at the age of 13. Are they correct? Let's call the school.

A last example is when in a univariate view everything is OK. The data is about the residents of a small city.

```
residents <- read_csv("residents.csv", col_types='i i l l l l')
```

```
summary(residents)
```

```
##      personID      age      employed      ownsHome  
## Min.   : 1.0    Min.   :18.00    Mode :logical    Mode :logical  
## 1st Qu.: 516.5  1st Qu.:31.00    FALSE:424        FALSE:837  
## Median :1032.0  Median :42.00    TRUE :1639       TRUE :1226  
## Mean   :1032.0  Mean   :42.22  
## 3rd Qu.:1547.5  3rd Qu.:54.00  
## Max.   :2063.0  Max.   :65.00  
  
## rentsHome      ownsCar  
## Mode :logical  Mode :logical  
## FALSE:1224     FALSE:635  
## TRUE :839      TRUE :1428  
##  
##  
##
```

Nothing weird. Or maybe: rents or owns home should be linked: who owns does not rent, who rent does not own.

Check them.

```

residents %>% filter(ownsHome==rentsHome)

## # A tibble: 6 × 6
##   personID   age employed ownsHome rentsHome ownsCar
##   <int> <int> <lgl>    <lgl>    <lgl>    <lgl>
## 1      17    55  TRUE    FALSE    FALSE    TRUE
## 2     134    19  TRUE     TRUE     TRUE    FALSE
## 3     203    43 FALSE     TRUE     TRUE    TRUE
## 4     382    60 FALSE     TRUE     TRUE    TRUE
## 5    1269    56  TRUE     TRUE     TRUE    TRUE
## 6    1902    31  TRUE    FALSE    FALSE    FALSE

```

Can you explain these 6 cases? What could happen?